

10 Dimensionality Reduction

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Large data



- Images: thousands or millions of pixels
- Documents: thousands of words, millions of letters
- Genomics: thousands of genes

Difficulties with high dimensionality

- Redundant/irrelevant features degrading the performance of (some) ML algorithms *eye color for grades*
- Difficult visualization in 3+ dimensions 
- Poor computational scaling $O(n^3)$ 
- Data sparsity (sparse = far from each other) causing difficulty finding patterns or clusters 
- Overfitting by capturing noise or irrelevant patterns
- Distance metrics lose meaning

How to reduce dimensions

- regularization* {
- Feature selection *ID, name*
 - Lasso (L1): drives coefficients to 0 (actual dimensionality)
 - Ridge (L2): drives coefficients close to 0 (effective dimensionality)
 - PCA (= principle component analysis)
 - t-SNE, autoencoders (MAL2)

Lossless vs. lossy compression

Lossless compression

- eliminating statistical redundancy
- all information can be re-constructed
- lower compression ratio (ex. 2:1)
- LZ77, LZ78, re-pair compression *Lempel Ziv*
- common formats: PNG, FLAC, ZIP, WAV

$(0,0,a)$
 $(0,0,b)$
 $(2,2,c)$
 $(2,3,a)$
 ababcbcbab

Lossy compression

- permanently removing "less important" information
- irreversible
- high compression ratio (ex. 50:1)
- PCA with dropping PCs with low eigenvalues
- common formats: JPEG, MP3, MP4

Example: lossless compression through linear combinations

Consider the following vectors:

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}, \begin{bmatrix} 5 \\ 10 \\ 15 \end{bmatrix}, \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix} \right\}$$

Storing each component in a byte, we need 18 bytes

$$= 6 \times 3$$

Example: lossless compression through linear combinations

$$\begin{aligned} 2\bar{v}_1 &= \bar{v}_2 \\ 4\bar{v}_1 &= \bar{v}_3 \\ 3\bar{v}_1 &= \bar{v}_4 \\ &\vdots \end{aligned}$$

$$\left\{ \begin{matrix} \bar{v}_1 \\ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \end{matrix}, \begin{matrix} \bar{v}_2 \\ \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \end{matrix}, \begin{matrix} \bar{v}_3 \\ \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix} \end{matrix}, \begin{matrix} \bar{v}_4 \\ \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} \end{matrix}, \begin{matrix} \bar{v}_5 \\ \begin{bmatrix} 5 \\ 10 \\ 15 \end{bmatrix} \end{matrix}, \begin{matrix} \bar{v}_6 \\ \begin{bmatrix} 6 \\ 12 \\ 18 \end{bmatrix} \end{matrix} \right\}$$

Using the original vector and multiplying constants, we only need

3 + 6 = 9

bytes.

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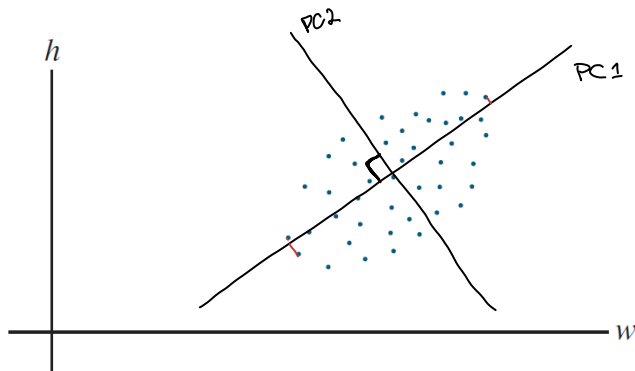
Visualization of PCA



Principle components (=PCs)

First PC is the projection direction that **maximizes the variance** of the projected data.

Second PC is the projection direction that is **orthogonal** to the first PC and **maximizes the variance** of the projected data.



PCA

PCA translates the columns into principle components. Each principle component is a **linear combination** of the features s.t. the PCs are **orthogonal** to each other (uncorrelated). If we have n features, n PCs explain 100% **variance** in data. PCs explain decreasing amount of variance with PC1 explaining the most variance.

$$PC_i = a_{i1}X_1 + a_{i2}X_2 + \dots + a_{in}X_n$$

We decrease the dimensionality of data by choosing $k < n$ number of PCs as a replacement to our original features.

PCA visualized

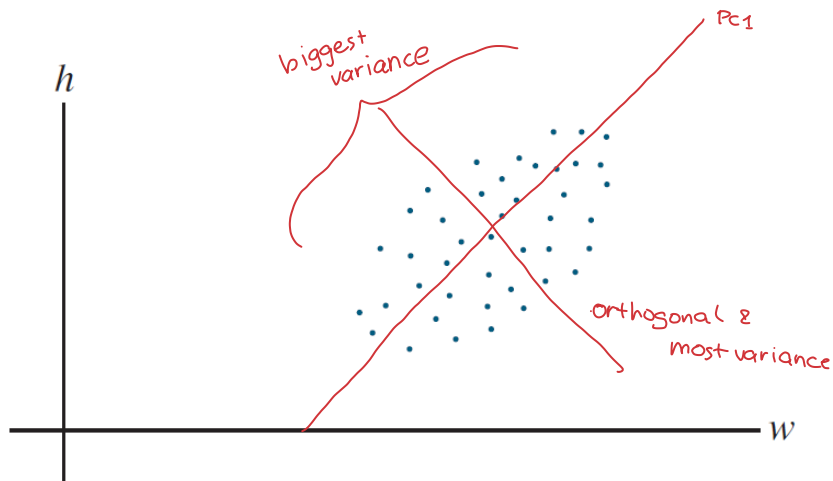


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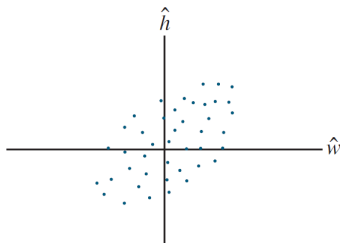
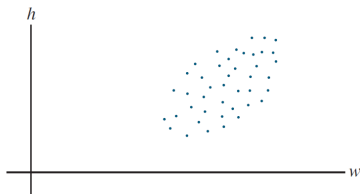
Math behind PCA

- 1 Translate data to mean-deviation form B (sample mean 0)
- 2 Find sample covariance matrix $S = \frac{1}{N-1}BB^T$
- 3 Find the eigenvectors and eigenvalues of S
- 4 $S = PDP^T$ where P consists of eigenvectors as columns and D is a diagonal matrix with eigenvalues on the diagonal, in a decreasing order.
- 5 Principle components $Y = P^T X$, linear combination of the features $x_1, \dots, x_p$ and the values of the eigenvector.

$$y_1 = \bar{u}_1^T \bar{X} = c_1 x_1 + c_2 x_2 + \dots + c_p x_p$$

Step 1: mean-deviation form B

Re-constructing the data so that they have 0 sample mean.



Step 1: mean-deviation form B

$\begin{bmatrix} \bar{X}_1 & \dots & \bar{X}_N \end{bmatrix}$ is a $p \times N$ observation matrix.

Sample mean:

$$M = \frac{1}{N}(\bar{X}_1 + \dots + \bar{X}_N)$$

The sample mean is the "center" of the scatter plot.

Step 1: mean-deviation form B

Translating all vectors by subtracting the sample mean

$$\hat{X}_k = \bar{X}_k - M$$

$$B = \begin{bmatrix} \hat{X}_1 & \hat{X}_2 & \dots & \hat{X}_N \end{bmatrix}$$

B is in **mean-deviation form**.

Example: sample mean

$$\bar{X}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \bar{X}_2 = \begin{bmatrix} 4 \\ 2 \\ 13 \end{bmatrix}, \bar{X}_3 = \begin{bmatrix} 7 \\ 8 \\ 1 \end{bmatrix}, \bar{X}_4 = \begin{bmatrix} 8 \\ 4 \\ 5 \end{bmatrix}$$

Sample mean:

$$M = \frac{1}{N}(\bar{X}_1 + \bar{X}_2 + \bar{X}_3 + \bar{X}_4)$$

$$M = \frac{1}{4} \left(\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \\ 13 \end{bmatrix} + \begin{bmatrix} 7 \\ 8 \\ 1 \end{bmatrix} + \begin{bmatrix} 8 \\ 4 \\ 5 \end{bmatrix} \right) = \begin{bmatrix} 5 \\ 4 \\ 5 \end{bmatrix}$$

Example: sample mean

Get B in mean-deviation form:

$$\hat{X}_k = \bar{X}_k - M$$

$$\hat{X}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ -4 \end{bmatrix}$$

$$B = [\hat{X}_1 \quad \hat{X}_2 \quad \hat{X}_3 \quad \hat{X}_4]$$

$$B = \begin{bmatrix} -4 & -1 & 2 & 3 \\ -2 & -2 & 4 & 0 \\ -4 & 8 & -4 & 0 \end{bmatrix}$$

$$\hat{X}_1 + \hat{X}_2 + \hat{X}_3 + \hat{X}_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Step 2: sample covariance matrix S

A (sample) **covariance matrix** is the $p \times p$ matrix:

$$S = \frac{1}{\underbrace{N-1}_{\text{degree of freedom}}} BB^T$$

Step 2: sample covariance matrix S



Diagonal: s_{jj} is the **variance** of x_j . Variance measures the spread of the values of x_j . The bigger the value, the wider the spread.

Total variance of the data is the sum of variances on the diagonal of S . The sum of diagonal entries of a square matrix S is called the **trace** of the matrix, written $tr(S)$.

The entry s_{ij} in S for $i \neq j$ is called the **covariance** of x_i and x_j . If the covariance is 0, x_i and x_j are **uncorrelated**. When S is a diagonal matrix, all entries are uncorrelated. S is always **symmetric** because $s_{ij} = s_{ji}$.

Example: covariance matrix

Sample covariance matrix:

$$S = \frac{1}{N-1} BB^T$$

Continuing the example from above, find S .

$$\begin{aligned}
 S &= \frac{1}{4-1} \begin{bmatrix} -4 & -1 & 2 & 3 \\ -2 & -2 & 4 & 0 \\ -4 & 8 & -4 & 0 \end{bmatrix} \begin{bmatrix} -4 & -2 & -4 \\ -1 & -2 & 8 \\ 2 & 4 & -4 \\ 3 & 0 & 0 \end{bmatrix} \\
 &= \frac{1}{3} \begin{bmatrix} 30 & 18 & 0 \\ 18 & 24 & -24 \\ 0 & -24 & 96 \end{bmatrix} = \begin{bmatrix} 10 & 6 & 0 \\ 6 & 8 & -8 \\ 0 & -8 & 32 \end{bmatrix}
 \end{aligned}$$

Example: covariance matrix

What is the variance of each feature? What does it mean?

$$\text{var}(x_1) = 10$$

x_3 has more spread
out data than

$$\text{var}(x_2) = 8$$

$$\text{var}(x_3) = 32$$

x_1 & x_2 .

$$\begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix} \begin{bmatrix} 10 & 6 & 0 \\ 6 & 8 & -8 \\ 0 & -8 & 32 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \end{matrix}$$

What is the trace?

$$\text{tr}(S) = 10 + 8 + 32 = 50$$

How are the features correlated?

$$\text{cor}(x_1, x_2) = 6$$

$\uparrow x_1 \quad \uparrow x_2$

house \$

build year

$$\text{cor}(x_1, x_3) = 0$$

uncorrelated

wall paint

$$\text{cor}(x_2, x_3) = -8$$

$\uparrow x_2 \quad \downarrow x_3$

distance to city

Step 3: eigenvectors and eigenvalues of S

eigenvector
eigenvalue

An **eigenvector** is a non-zero vector $\bar{x}$ such that $A\bar{x} = \lambda\bar{x}$, $\lambda \in \mathbb{R}$.

A scalar λ is an **eigenvalue** of A if there is a non-trivial solution $\bar{x}$ of $A\bar{x} = \lambda\bar{x}$.

In such cases we say that $\bar{x}$ is an eigenvector corresponding to λ .

$T : \bar{x} \rightarrow A\bar{x}$.

λ is an eigenvalue of A iff $A - \lambda I = 0$ has a non-trivial solution.

Example: eigenvectors

Are $\bar{u}$ and $\bar{v}$ eigenvectors of A ?

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}, \bar{u} = \begin{bmatrix} -12 \\ 10 \end{bmatrix}, \bar{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$A\bar{x} = \lambda\bar{x}$$

$$A \cdot \bar{u} = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} -12 \\ 10 \end{bmatrix} = \begin{bmatrix} -12 + 60 \\ -60 + 20 \end{bmatrix} = \begin{bmatrix} 48 \\ -40 \end{bmatrix} = -4 \cdot \begin{bmatrix} -12 \\ 10 \end{bmatrix}$$

$\rightarrow \bar{u}$ is an eigenvector

$$A \cdot \bar{v} = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 3 - 12 \\ 15 - 4 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix}$$

$\rightarrow \bar{v}$ is NOT an eigenvector

Example: eigenvalues

$$A\bar{x} = \lambda\bar{x}$$

Determine if $\lambda = 7$ is an eigenvalue of A :

$$A\bar{x} = 7\bar{x} \implies A\bar{x} - 7I\bar{x} = 0 \implies (A - 7I)\bar{x} = 0$$

identity matrix

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - 7 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-7 & 6 \\ 5 & 2-7 \end{bmatrix} = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$$\text{PVF: } \bar{x} = x_2 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

↑ eigenvector

↑ free var.

→ non-trivial sol.

→ $\lambda = 7$ is an eigenval.

The characteristic equation

A scalar λ is an eigenvalue of an $n \times n$ matrix iff λ satisfies

$$\det(A - \lambda I) = 0$$

where $\det$ is the **determinant** of the matrix.

Determinant for a 2x2 matrix, recap from MSE

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, A is invertible.

$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

To calculate the determinant of a bigger matrix, we would need to perform cofactor expansion - again, to be continued in AL1 :)

Note: you can use Python to help with calculations, see lesson 2 Github repo.

Example: eigenvalues

Find eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$

$a d - b c$

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{vmatrix} = (2-\lambda)(-6-\lambda) - 3 \cdot 3 = -12 - 2\lambda + 6\lambda + \lambda^2 - 9$$

$$= \lambda^2 + 4\lambda - 21$$

$$0 = (\lambda + 7)(\lambda - 3)$$

$$\lambda_1 = -7$$

$$\lambda_2 = 3$$

eigenvalues of A

$$\{-7, 3\}$$

Step 4: eigenvectors and eigenvalues of S

$$A = U \cdot \Sigma \cdot V^T$$

$$S = PDP^T$$

- S : covariance matrix,
- D : diagonal matrix with eigenvalues of S on the diagonal in decreasing order,
- P : orthogonal matrix with columns as unit eigenvectors $\bar{u}_1, \dots, \bar{u}_p$ corresponding to the eigenvalues.

singular value decomposition

This comes from SVD. Background and proof to be continued in ALI :)

Example from before continued

$$S = \begin{bmatrix} 10 & 6 & 0 \\ 6 & 8 & -8 \\ 0 & -8 & 32 \end{bmatrix}$$

Construct the eigen-decomposition $S = PDP^T$. Use Python to find the matrices.

CODE

Step 5: Find principle components

The unit vectors $\bar{u}_1, \dots, \bar{u}_p$ of the covariance matrix S are called the **principal components** of the data. The first principal component is the eigenvector corresponding to the largest eigenvalue of S .

The first principal component $\bar{u}_1$ determines a new variable y_1 :

$$y_1 = \bar{u}_1^T \bar{X} = c_1 x_1 + c_2 x_2 + \dots + c_p x_p$$

following the equation $Y = P^T X$. $c_1, \dots, c_p$ are entries in $\bar{u}_1$. y_1 is a linear combination of the original variables $x_1, \dots, x_p$, using the eigenvector $\bar{u}_1$ as weights.

Example: find the PCs

Find the principle components and define them as linear combinations of the features.

$$P = [\bar{v}_1 \quad \bar{v}_2 \quad \bar{v}_3]$$

$$\bar{v}_1 = \begin{bmatrix} a_1 \\ b_1 \\ c_1 \end{bmatrix}$$

$$P^T = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$PC\ 1 = \underline{a_1}x_1 + \underline{b_1}x_2 + \underline{c_1}x_3$$

$$PC\ 2 = a_2x_1 + b_2x_2 + c_2x_3$$

$$PC\ 3 = a_3x_1 + b_3x_3 + c_3x_3$$

Finally: using PCs

10:57 - 11:05

Given n features, they are translated into n PCs. All n PCs capture 100% of variance in the data. PC1 captures the most variance, PC2 less, etc.

The variance of each PC is λ_i and $\lambda_i / \text{tr}(S)$ measures the fraction of total variance that is explained (captured) by the given PC.

λ
 Lambda

Example: explained variance

diagonal m.

$$D = \begin{bmatrix} 34.55 & 0 & 0 \\ 0 & 13.84 & 0 \\ 0 & 0 & 1.61 \end{bmatrix} \leftarrow \begin{array}{l} \text{ordered eigenvalues} \\ \text{decreasing} \end{array}$$

Compute the trace and how much variance each of the PCs explain.

$$\text{tr}(S) = 34.55 + 13.84 + 1.61 = 50.00$$

$$\left. \begin{array}{l} \text{PC 1: } \frac{34.55}{50} = 0.691 \\ \text{PC 2: } \frac{13.84}{50} = 0.277 \\ \text{PC 3: } \frac{1.61}{50} = 0.032 \end{array} \right\} \Sigma = 1$$

$$S = \begin{bmatrix} 10 & 6 & 6 \\ 6 & 8 & -8 \\ 0 & -8 & 32 \end{bmatrix}$$

trace is invariant
under orthogonal
rotation

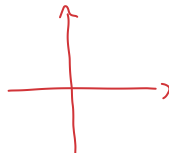
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PCA for dimensionality reduction

PCA on its own is just an **analysis** and does not reduce dimensions.
To reduce dimensions:

- Choose PCs s.t. certain percentage of variance is captured (e.g. 80%)
- Choose only PC1 and PC2 (visualization in 2D)
- Elbow/shoulder method



Elbow method

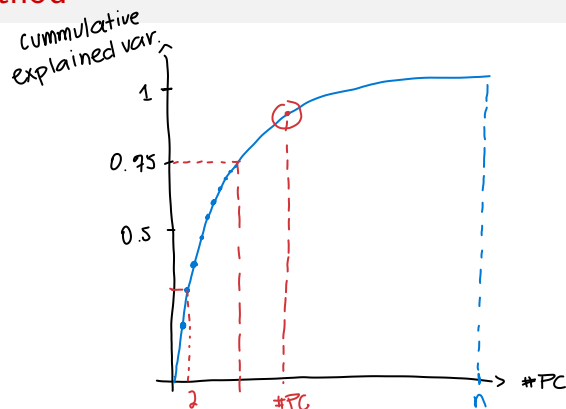


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Code

Github link: https://github.com/304125/MAL1_S26/tree/master/10_Dimensionality_Reduction